

Element F: Consideration of Design Viability

Project Title: MJ's LAST Stand (Echo Rack)

Part 1: Identify Possible Risks

Risk #	What Could Go Wrong?
1.	The gong stand is not able to hold the gong.
2.	The gong is built poorly and therefore unsafe for normal use.
3.	The gong stand is way over budget. (at least \$25 over)
4.	The mute for the gong stand does not work.
5.	The wheels do not work properly.

Part 2: Group the Risks

Risk #	What Could Go Wrong?	Risk Type
1.	The gong stand is not able to hold the gong.	Operational
2.	The gong is built poorly and therefore unsafe for normal use.	Operational
3.	The gong stand is way over budget. (at least \$25 over)	Financial
4.	The mute for the gong stand does not work.	Operational
5.	The gong is not able to be transported.	Operational

Part 3: How Likely Are These Risks?

Risk #	How Likely is the Risk? (L,M,H)	How Bad Would It Be? (L,M,H)	Priority
1	Low	High	First
2	Low	High	Third

3	High	Medium	Second
4	Medium	Low	Fifth
5	Low	Medium	Fourth

Part 4: How to Prevent These Risks

Risk	How To Stop Or Fix This Risk?	Why Will This Work?
#1	Avoid it: The possibility of the gong stand not being able to hold the gong is extremely low therefore avoiding the risk wouldn't be bad.	Again, the possibility of the gong stand not being able to hold the gong is very unlikely. I'd be willing to bet money on it. There is no reason we should really worry about it not working so avoiding it might be optimal.
#2	Reduce the Risk: Use proper safety standards to construct the gong, and go through extensive testing with the finished product.	If we build the gong stand and test it extensively, it is unlikely that an issue will present itself that wasn't a problem during testing.
#3	Reduce the Risk: Use cheap and cost effective materials, maybe order in bulk.	Ordering in bulk typically reduces the price for items because of how much you're ordering, so that could be a way for us to reduce the price on materials like wood.
#4	Avoid it: If unfortunately we can't get the mute design to work, we may choose to scrap the mute idea of the project as it is not the main priority of the project. Of course, we will try to get it working first as it really comes down to design and build quality.	There's a good chance in both directions, that the mute will or won't work. Basically reiterating what was stated before, the mute is the novel part of the design and is not necessary to the operation of the gong stand itself.
#5	Reduce the Risk: Assure that the wheels are working through extensive testing, also oil them.	Oiling the wheels will prevent rust and built up/dirt from affecting its movement capabilities.

Part 5: Keeping an Eye on Things

Risk #	How Will You Keep An Eye On It?	How Will You Tell The Team?
1	Check the structure of the gong stand making sure the base is sturdy enough to hold the gong.	As a group, it is our responsibility to look out for each other. If one of us sees something wrong with our structure, they should speak up for the betterment of our team. This should apply to all of our efforts as well, not just the structure.
2	Use proper safety equipment and manage tools safely and responsibility. Do not play around with equipment.	Again, it is our responsibility to look out for each other and hold ourselves accountable for our actions. If we see each other messing around, it is our job to tell others to stop. This also means that if someone is more responsible than someone else, it might be more beneficial to have them use the tools versus someone else.
3	Take note of how much has been spent so far, every now and then, to better understand the cost of the project.	Discuss with members on the cost of materials and on decisions of what materials will be purchased.
4	As the project progresses, continue to consider the validity of the gong mute.	Hold frequent discussions with the team on the issue.
5	Test the gongs wheels to see if it is able to move and be transported.	Discuss any problems or any concerns regarding the transportation of the gong.

Part 6: Reflection

Instructions: Document your responses to the reflection questions below in paragraph format.

1. Which risk do you think is the most important? Why?

The most important risk out of the five we have considered is Risk #1: The stand being incapable of securely holding the gong. This risk bears the highest priority as it would basically determine whether the project is a success or failure (a gong stand that can't hold a gong is useless), however, the likelihood of this risk is low. Nevertheless, strong effort will be undertaken to limit the possibility of such a thing occurring.

2. How did you decide on the ways to stop the risks? Was there a specific method you liked best?

Holding ourselves accountable for ourselves and our group mates is an excellent way of making sure errors are minimal, and we will also be testing after installing each component of the device, so that when the final device is constructed with all tested components installed, it would be extremely unlikely for a previously mentioned error to occur.

3. How will you make sure your plans stay on track?

As a team, we will hold each other accountable to ensure we are aware of these risks and the plans to mitigate them. We will hold each other accountable if they are not followed.